

Concept Review: Integration by Parts

Name: _____

Learning goals. By the end of recitation, you should be able to choose appropriate expressions for u and dv , apply integration by parts to indefinite and definite integrals, and check an antiderivative by differentiating.

Where does the formula come from?

Integration by parts reverses the _____ rule:

$$\frac{d}{dx}(u(x)v(x)) = \text{_____} + \text{_____}.$$

Integrating and rearranging gives

$$\boxed{\int u \, dv = uv - \int v \, du.}$$

The goal is to choose u and dv so that the new integral $\int v \, du$ is *easier* than the original integral.

What is different for the definite integral:

$$\int_a^b f(x)g'(x)dx = \text{_____} + \text{_____}.$$

How do we choose u and dv ?

- Choose u to be a factor that becomes simpler when differentiated.
- Choose dv to be a factor that you know how to integrate.

A common guideline for choosing u is **ILATE**:

- I Inverse trigonometric functions, such as $\arctan x$
- L Logarithmic functions, such as $\ln x$
- A Algebraic functions, such as x or x^2
- T Trigonometric functions, such as $\sin x$
- E Exponential functions, such as e^x

The idea is that when an integrand contains two types of functions, you generally choose as u whichever appears earlier in the list.

ILATE is a guideline, not a theorem. Always ask whether the resulting integral is actually simpler.

Quick concept check

For

$$\int x e^{3x} dx,$$

choose u and dv , and then find du and v .

$$u = \underline{\hspace{2cm}} \qquad dv = \underline{\hspace{2cm}}$$

$$du = \underline{\hspace{2cm}} \qquad v = \underline{\hspace{2cm}}$$

Why does this choice make the new integral easier?

Reminders

Before applying integration by parts, ask yourself:

- Which part of the integrand becomes simpler when differentiated?
- Can I easily find an antiderivative of the part chosen as dv ?
- Will the new integral $\int v du$ be easier than the original integral?
- Would another method, such as substitution, be more efficient?

After applying integration by parts, check:

- Did I calculate both du and v correctly, including all constants and signs?
- Did I correctly use the subtraction in $\int u dv = uv - \int v du$?
- Did I include $+C$ for an indefinite integral or correctly evaluate the bounds for a definite integral?
- Does differentiating my answer reproduce the original integrand?